Part 1:

a. Software by itself (left without supervision) especially in health, security or weaponry situations is unsafe because humans create software and all humans are imperfect. Because we are imperfect the software we create to be used in centers like health or security are bound to have mistakes happen and some could cause injury or even worse be fatal. Software safety solely depends on the context where the software is used.
b. It should come in from the beginning of software development because if it doesn't safety cannot be assured safety should be the priority over assurance of the software.
c. The safety of software depends on the context in which it is used. Some software can be reused and would give no issues while some can be reused and later cause issues that can be fatal or unsafe because they have no error handling error checking or any software safety measures.
d. I think using Object Oriented Technology can lead to safer software because it would involve keeping undercover errors that could lead to an accident. It also allows reusing of software and maintenance, it helps to provide some security to the software and operator.
e. If we are solely looking at just the safety of software and looking away from the main functionality of the software, it would be safer first to implement the error-handler to cover possible scenarios of how the software could fail or cause issues, then implement safety measures for them, then the normal behaviour can be implemented.

Part 2

a. Name: INSTALLING RAILS(Use case 1)
Primary Actor(s): Technician
Stakeholder and Interests: Technician - Wants to install rail brackets and guiding rails for the elevator.
Precondition(s): Technicians must be capable of doing the installments and must have the right equipment needed. The elevator Shaft must also be properly framed to accommodate the car sling.
Success Guarantee(s): The upward and downward motion of the elevator causes no issues.
Main Success Scenario:

1. Install spot brackets at the top part of the Elevator Shaft.
2. technicians drop a steel wire Plumb line into the elevator pit to line the lower brackets with the top shaft Spots.
3. Technicians attach a guide rail using a ton chain hoist and install position magnets which helps the elevator to stop on the right floor.
4. Install motion control system used to temporarily move the elevator while on construction.
5. install car sling used to carry passengers.

Extensions:

1. improper installment of sports brackets wow which can cause misalignments of the Plumb line to the lower bracket.

2. improper placement of guide rail, faulty sensors which could cause issues with the elevator stopping on the wrong floor.
3. Inability to install car sling due to improper measurements of the shaft.

Name: INSTALLING CAR SLING, CAB AND DOOR(Use case 2)
Primary Actor(s): Technician
Stakeholder and Interests: Technician - wants to install car sling, cab, and doors.
Precondition(s): Use case 1 is completed and fully functional.
Success guarantees: The elevator does a proper opening and closing in sync, and the cab is properly installed without issues.
Main Success Scenario:

1. Install Struts and brackets to hold the door securely.
2. Place the hoistway sill on each level properly aligning it with the capsule.
3. Place headers along the top entrances to complete the frame of the door.
4. Place the door on the landing header to rest on the smooth rolling caster and keep them on track using Gibbs.
5. Assemble the side and rear interior walls along the dome and ceiling units.

Extensions:

1. Improper placement of struts and brackets.
2. Misaligned hoist way sill.
3. The side and rear interior walls are not properly fitted in.

Name: WIRING(use case 3)
Primary Actor(s): Electrician
Stakeholder and Interests: electrician - Wants to provide computation services to the elevator.
Precondition(s): Use cases 1 and 2 are completed and fully functional.
Success guarantees: A fully functional automated elevator.
Main Success Scenario:

1. Fieldwire car control system while removing jumpers that override computer control.
2. remove the run box used to control the elevator during installation.
3. Install sensors for reopening doors when instructed.
4. Electrical wiring for lights and ventilation fan.

Extensions:

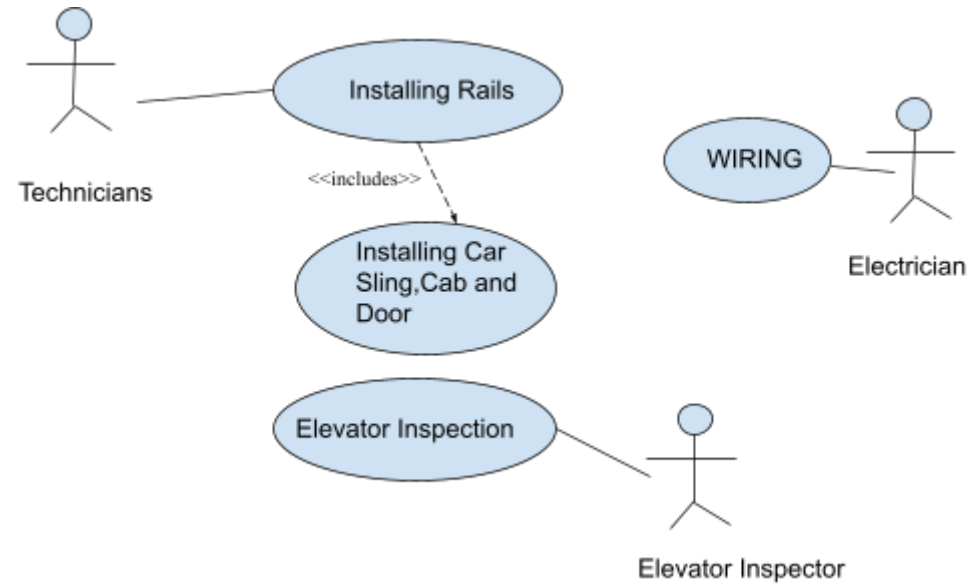
Incorrect wire connections
and faulty equipment used which could cause the elevator to not function properly.
Sensors not properly detecting the correct floor to stop.

Name: INSPECTION (use case 4)
Primary Actor(s): Elevator inspector
Stakeholder and Interests: Elevator inspector- Wants to ensure complete functionality of the elevator and working safety measures.
Precondition(s): Use cases 1,2 and 3 are completed and fully functional.
Success guarantees: The elevator is fully functional and ready to be used by passengers
Main Success Scenario:
Adjust the door latches and correct the sensors if needed,validate and ensure that the elevator moves at the correct speed; if not adjust the valve that controls the rate at which hydraulic fluid is pumped in and out of the Pistons until the speed meets specifications while testing to make sure the buttons and switches are fully connected and functional. Lastly,ensure the door and fire safety operations are fully functional.

Extensions:

1. faulty sensors needing a replacement.
2. elevator moves faster or slower than the required speed.
3. Hydraulic fluid is not being pumped properly.
4. Buttons and switches not doing what is instructed.
5. elevator does not apply safety measures.

b.



PART 3

a. Name: HELP
Primary Actor(s): User, Emergency Services, Building Safety Service
Stakeholder and Interests: User - wants to be request help using the button in the elevator,Emergency Services - Wants to send help as needed to the user requested, Building Safety Service- wants to alert/provide help to the user as requested.
Precondition(s): Elevator must be fully functional/Operational.
Success Guarantee(s): When the button is pressed by a user, an alarm signal should be sent to the control system signaling that help was requested.
Main Success Scenario:

1. User presses help button to indicate that help is requested.
2. The control system receives a “Help” alarm signal from an elevator indicating that the “Help” button has been pressed.
3. The passenger is connected to building safety service through a voice connection.
4. If there is no response from building safety within 5 seconds or if there is no response from a passenger a 911 emergency call is placed.

Extensions:

1. Help button doesn’t send a signal/the appropriate signal to the control system when pressed.

Name: DOOR OBSTACLES
Primary Actor(s): Building Safety Service
Stakeholder and Interests: Building Safety Service - wants to ensure that the light sensor properly works during the doors peoning and closure.
Precondition(s): Elevator must be fully functional/Operational.
Success Guarantee(s): If the light sensor is interrupted when the door is closing, the control system stops the door from closing and opens it. If this occurs repeatedly over a short period of time, a warning is sounded over the audio system and a text message is displayed.
Main Success Scenario:

1. When the door is closing, test the sensor by attempting to walk in.

Extensions:

1. Light sensor not properly detecting obstacles.

Name: FIRE
Primary Actor(s): User, Building Safety Service
Stakeholder and Interests: User - wants to send a signal to the control system about a fire emergency, Building Safety Service - wants to accurate respond to the fire signal being received.
Precondition(s): Elevator must be fully functional/Operational.
Success Guarantee(s): The control system receives a “Fire” alarm signal from the building and commands all elevators to move to a safe floor. Similarly, a “Fire” alarm signal from the elevator itself will cause that elevator to go to a safe floor. In both cases an audio and text message are presented to passengers informing them of an emergency and asking them to disembark once the safe floor is reached.
Main Success Scenario:

1. User presses the fire alarm button, sending a sigal to the control system which makes all elevators move to a safe floor.

Extensions:

1. Fire alarm button doesn't send a signal/the appropriate signal to the control system when pressed.
2. Elevators get stuck/does not move to a safe floor.

Name: OVERLOAD
Primary Actor(s): Building Safety Service
Stakeholder and Interests:Building Safety Service - wants to ensure the passengers maintain the required elevator load.
Precondition(s): Elevator must be fully functional/Operational.
Success Guarantee(s): The control system receives an "Overload" alarm signal from an elevator if the sensors indicate that the passenger or cargo load exceeds the carrying capacity. In that case, the elevator does not move and an audio and a text messages are presented to passengers asking for the load to be reduced before attempting to move again.
Main Success Scenario:

1. The control system receives an "Overload" alarm signal from an elevator.
2. The sensors indicate that the passenger or cargo load exceeds the carrying capacity.
3. At this stage, the elevator does not move and an audio and a text messages should be sent to passengers asking for the load to be reduced before attempting to move again.

Extensions:

1. Elevator does not inform passengers of its required passenger load.
2. Elevator becomes unstable but still moves upward and downward.

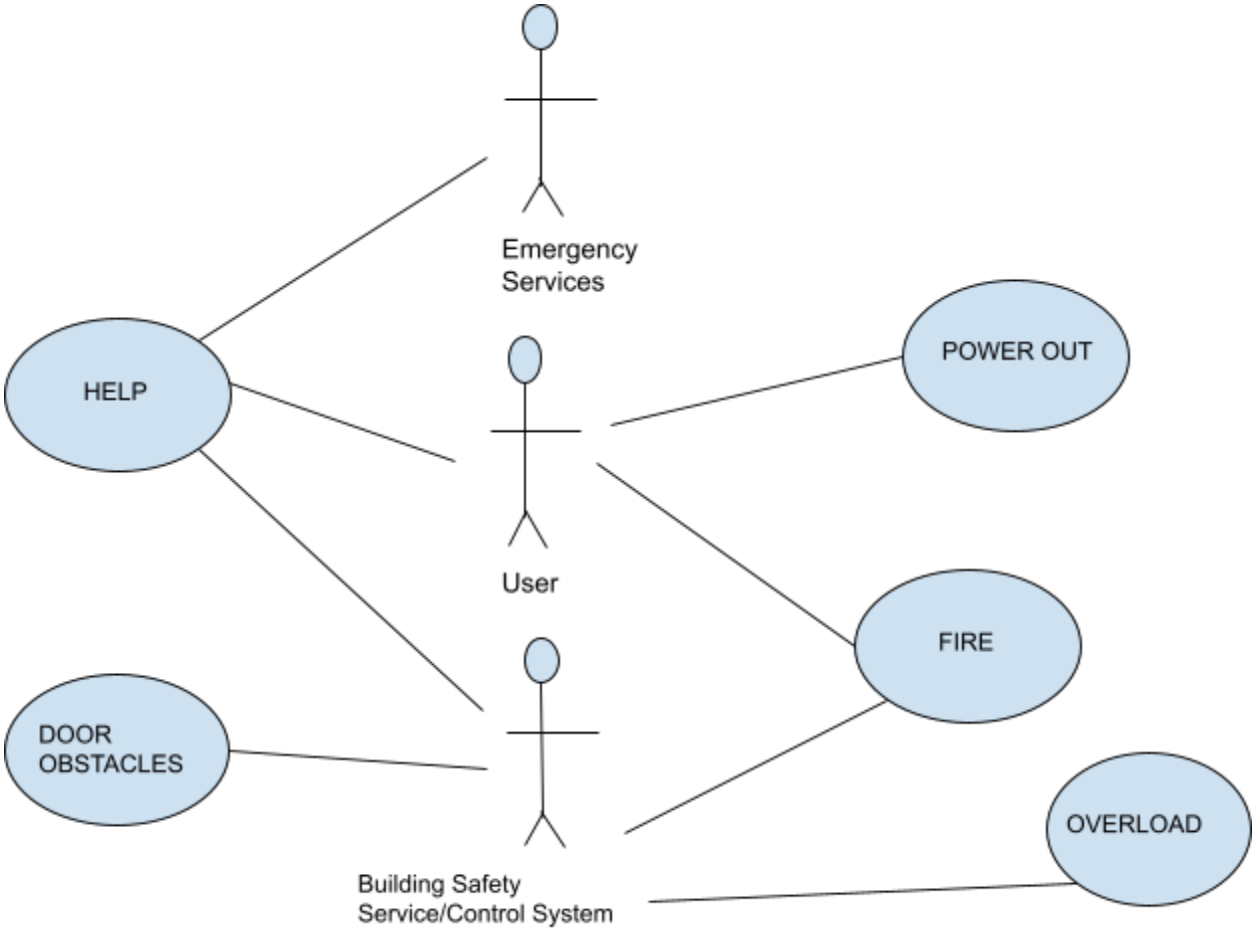
Name: POWER OUT
Primary Actor(s): User
Stakeholder and Interests: User - wants to ensure the elevator moves to a safe floor in a power outage event.
Precondition(s): Elevator must be fully functional/Operational.
Success Guarantee(s): The control system receives a "Power Out" alarm signal. In that case, an audio and a text messages are presented to passengers informing them of the power outage. Each elevator is then moved to a safe floor and passengers are asked to disembark via audio and text messages. The battery backup power is sufficient to do all of this.
Main Success Scenario:

1. .Passengers should receive info about power outage and instructions on how to proceed via audio or text message.
2. Elevators should move passengers to a safe floor

Extensions:

1. Passengers are unable to receive instructions via audio or text message.
2. Elevator is unable to move or drop passengers on a safe floor.

b.


Emergency
Services
HELP
POWER OUT
User
FIRE
DOOR
OBSTACLES
OVERLOAD
Building Safety
Service/Control System